Functions, Modulus and Quadratics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises**

The line $y = mx - 1$ intersects the curve $y = x^2 + 3x + k$ at two distinct points for all real values of m. Find the range of values of the constant k.

To find the points of intersection of the line and the curve, set their equations equal: $x^2 + 3x + k = mx - 1$ Rearranging into standard quadratic form $ax^2 + bx + c = 0$: $x^2 + (3 - m)x + (k + 1) = 0$ For the line and curve to intersect at two distinct points, the discriminant with respect to x, denoted Δ_x, must be strictly positive: $\Delta_x = (3 - m)^2 - 4(1)(k + 1) > 0$ Expanding and simplifying: $\Delta_x = m^2 - 6m + 9 - 4k - 4 = m^2 - 6m + (5 - 4k) > 0$ We are given that this inequality holds.

2 **[Foundational] • Target: Function Mapping, Domain/Range and Inverses**

The line $y = 7x - 4$ is tangent to the curve $y = ax^2 + \frac{b}{x}$ at the point where $x = 2$, where a and b are constants and $xe0$. Find the coordinates of the stationary point of the curve and determine its nature.

1. Find the point of tangency: Substitute $x = 2$ into the tangent line equation $y = 7x - 4$: $y = 7(2) - 4 = 10$ Since the line touches the curve at $x = 2$, the point $(2, 10)$ lies on the curve $y = ax^2 + \frac{b}{x}$: $a(2)^2 + \frac{b}{2} = 10 \Rightarrow 4a + \frac{b}{2} = 10 \Rightarrow 8a + b = 20 \quad - - - \ (1)$ 2. Use the gradient of the tangent: Differentiating $y = ax^2 + bx^{-1}$ gives: $\frac{dy}{dx} = 2ax - \frac{b}{x^2}$ The gradient of the tangent line $y = 7x - 4$ is 7. At $x = 2$.

3 **[Foundational] • Target: Modulus Equations and Inequalities**

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.

To find the area of the enclosed region, we synthesize differential calculus, polynomial algebra, and definite integration: 1. Find the equation of the tangent line at $P(2, 4)$: Differentiating the curve equation: $\frac{dy}{dx} = 3x^2 - 3$ At $x = 2$, the gradient of the tangent is: $m = 3(2)^2 - 3 = 12 - 3 = 9$ Using the point-slope formula with $P(2, 4)$: $y - 4 = 9(x - 2) \Rightarrow y = 9x - 14$ 2. Find the intersection point Q: Set the curve and tangent equations equal.

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4 [Foundational] • Target: Quadratic Discriminant and Line-Parabola Intersections

A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.

Step 1: Find the points of intersection of the curve and the line. Set the equations equal to each other: $6x - x^2 = mx \Rightarrow x^2 - (6 - m)x = 0 \Rightarrow x(x - (6 - m)) = 0$ Since the line intersects the curve at the origin $(0, 0)$ and at point P, the x-coordinate of P is $x = 6 - m$. Step 2: Set up the definite integral for the enclosed area. For $0 \le x \le 6 - m$, the curve lies above the straight line. The enclosed area A is given by.

5 [Foundational] • Target: Completing the Square and Applied Optimization

Find the equation of the tangent line to the curve $y = (4x + 1)^3$ at the point where $x = 0$.

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(4x + 1)^2 \cdot (4) = 12(4x + 1)^2$. At $x = 0$, slope $m = 12$. Equation: $y - 1 = 12(x - 0) \Rightarrow y = 12x + 1$. Final Answer: $y = 12x + 1$

6 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The variables x and y satisfy the relationship $y = Ax^n$, where A and n are constants. When $\log_2 y$ is plotted against $\log_2 x$, a straight line is obtained which passes through the points $(1, 5)$ and $(4, -1)$. What is the value of y when $x = 4$?

1. Express the non-linear relationship in linear form $Y = mX + c$: Taking base-2 logarithms of both sides of $y = Ax^n$: $\log_2 y = \log_2 (Ax^n) = n\log_2 x + \log_2 A$ Letting $Y = \log_2 y$ and $X = \log_2 x$, the equation becomes: $Y = nX + \log_2 A$ where the gradient is $m = n$ and the vertical intercept is $c = \log_2 A$. 2. Find the equation of the straight line: Using the given points $(X_1, Y_1) = (1, 5)$ and $(X_2, Y_2) = (4, -1)$: $m = \frac{-1-5}{4-1} = \frac{-6}{3} = -2$ Using the point.

7 [Foundational] • Target: Function Mapping, Domain/Range and Inverses

Variables x and y satisfy the equation $y = Ax^k$, where A and k are constants, with $x > 0$. The graph of $\ln y$ against $\ln x$ is a straight line passing through the points $(\ln 2, \ln 72)$ and $(\ln 6, \ln 8)$. Find the value of x when $y = 18$.

Step 1: Reduce the non-linear relationship to linear form. Taking the natural logarithm of both sides of $y = Ax^k$: $\ln y = \ln(Ax^k) = \ln A + k\ln x$ This is in the linear form $Y = mX + c$, where $Y = \ln y$, $X = \ln x$, gradient $m = k$, and vertical intercept $c = \ln A$. Step 2: Determine the gradient k. Using the given points $(\ln 2, \ln 72)$ and $(\ln 6, \ln 8)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Modulus Equations and Inequalities

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 18$. The particle is at instantaneous rest when $t = 1$. Given that the displacement of the particle from a fixed point O is $s = -10$ m when $t = 2$, find the displacement of the particle, in metres, when it is next at instantaneous rest.

1. Find the velocity function $v(t)$: $v(t) = \int a\,dt = \int (6t - 18)\,dt = 3t^2 - 18t + c$ Since the particle is at instantaneous rest when $t = 1$, $v(1) = 0$: $3(1)^2 - 18(1) + c = 0 \Rightarrow -15 + c = 0 \Rightarrow c = 15$ Thus, $v(t) = 3t^2 - 18t + 15$. 2. Find when the particle is next at instantaneous rest: Set $v(t) = 0$: $3(t^2 - 6t + 5) = 0 \Rightarrow 3(t - 1)(t - 5) = 0$ So the particle is at rest at $t = 1$ and next at $t = 5$. 3. Find the displacement function $s(t)$.

9 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

A particle moves in a straight line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 6t^2 - 18t + k$, where k is a constant. Given that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first 2 seconds.

Step 1: Determine the unknown parameter k. The particle is momentarily at rest when $t = 2$, so $v(2) = 0$: $6(2)^2 - 18(2) + k = 0 \Rightarrow 24 - 36 + k = 0 \Rightarrow k = 12$ Step 2: Find all times when the particle is at rest to check for direction changes. $v(t) = 6t^2 - 18t + 12 = 6(t^2 - 3t + 2) = 6(t - 1)(t - 2) = 0$ The particle is at rest at $t = 1$ and $t = 2$. In the interval $0 \le t < 1$, $v(t) > 0$; in the interval $1 < t < 2$, $v(t) < 0$. Step 3: Calculate the total distance traveled by integra.

10 [Intermediate] • Target: Completing the Square and Applied Optimization

The quadratic equation $x^2 - (2k + 1)x + (k^2 + 2) = 0$ has real roots α and β. Given that $\alpha^2 + \beta^2 = 27$, find the value of the constant k.

From the relations between roots and coefficients for the quadratic equation $x^2 - (2k + 1)x + (k^2 + 2) = 0$: $\alpha + \beta = 2k + 1 \quad \text{and} \quad \alpha\beta = k^2 + 2$ Expressing $\alpha^2 + \beta^2$ in terms of the sum and product of the roots: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ Substitute the known values: $(2k + 1)^2 - 2(k^2 + 2) = 27$ Expand and simplify: $(4k^2 + 4k + 1) - (2k^2 + 4) = 27$ $2k^2 + 4k - 3 = 27 \Rightarrow 2k^2 + 4k - 30 = 0 \Rightarrow k^2 + 2k - 15 = 0$ F.

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11 **[Intermediate]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

The variables x and y satisfy the equation $y = Ab^x$, where A and b are positive constants. When $\log_2 y$ is plotted against x, a straight line is obtained that passes through the points $(2, 7)$ and $(5, 13)$. Find the value of y when $x = 4$.

To reduce the non-linear relationship $y = Ab^x$ to linear form, take base-2 logarithms of both sides: $\log_2 y = \log_2 (Ab^x) = \log_2 A + x\log_2 b$ Setting $Y = \log_2 y$ and $X = x$, the equation represents a straight line $Y = mX + c$, where the gradient is $m = \log_2 b$ and the vertical intercept is $c = \log_2 A$. 1. Find the gradient m: $m = \frac{13-7}{5-2} = \frac{6}{3} = 2$ 2. Find the vertical intercept c using the point $(2, 7)$: $7 = 2(2) + c \Rightarrow c = 3$ Thus, the linear equation i.

12 **[Intermediate]** • **Target: Function Mapping, Domain/Range and Inverses**

The variables x and y satisfy the relationship $y = Ab^x$, where A and b are positive constants. The graph of $\ln y$ against x is a straight line passing through the points $(0, \ln 3)$ and $(4, \ln 48)$. What is the value of b?

Taking the natural logarithm of both sides of the equation $y = Ab^x$: $\ln y = \ln(Ab^x) = \ln A + x\ln b$ This is in linear form $Y = mX + c$, where $Y = \ln y$, $X = x$, the gradient is $m = \ln b$, and the vertical intercept is $c = \ln A$. From the point $(0, \ln 3)$, the intercept is $\ln A = \ln 3$, which gives $A = 3$. Using the second point $(4, \ln 48)$: $\ln 48 = \ln 3 + 4\ln b$ Subtract $\ln 3$ from both sides and apply the quotient rule of logarithms.

13 **[Intermediate]** • **Target: Modulus Equations and Inequalities**

The function f is defined for $xeq3$ by $f(x) = \frac{2x+1}{x-3}$. Find an expression for $f^{-1}(x)$.

To find the inverse function $f^{-1}(x)$, let $y = f(x)$: $y = \frac{2x+1}{x-3}$ Multiply both sides by $(x - 3)$: $y(x - 3) = 2x + 1$ Expand the left-hand side: $xy - 3y = 2x + 1$ Rearrange to collect all terms involving x on one side: $xy - 2x = 3y + 1$ Factor out x: $x(y - 2) = 3y + 1$ Divide by $(y - 2)$: $x = \frac{3y+1}{y-2}$ Replacing y with x gives: $f^{-1}(x) = \frac{3x+1}{x-2}, \quad xeq2$. Common misconceptions: - $\frac{x-3}{2x+1}$ confuses the invers.

14 **[Intermediate]** • **Target: Quadratic Discriminant and Line-Parabola Intersections**

The function f is defined for all real numbers x by $f(x) = 2 - 3\sin(4x)$. What are the amplitude and the period (in radians) of f?

For a sinusoidal function written in the form $f(x) = c + a\sin(bx)$ (or $c - a\sin(bx)$): 1. Amplitude: The amplitude represents the maximum vertical distance from the horizontal centerline (midline $y = 2$) to a peak or trough. It is defined as the absolute value of the coefficient of the trigonometric ratio: $\text{Amplitude} = |-3| = 3$. Amplitude is strictly a non-negative measure of distance; it can never be negative. 2. Period: The fundamental period of $\sin(x)$ is 2π. For $\sin(bx)$, the fun.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities**

The functions f and g are defined for all real values of x by $f(x) = 2x - 1$ and $g(x) = \frac{x+5}{3}$. Find the value of x such that $fg(x) = 9$.

To find x such that $fg(x) = 9$, we can either determine the composite expression $fg(x)$ or solve sequentially from the outer function to the inner function. Method 1 (Sequential Evaluation): We have $f(g(x)) = 9$. Since $f(u) = 2u - 1$, setting $f(u) = 9$ gives: $2u - 1 = 9 \Rightarrow 2u = 10 \Rightarrow u = 5$. Thus, $g(x) = 5$. Now solve for x: $\frac{x+5}{3} = 5 \Rightarrow x + 5 = 15 \Rightarrow x = 10$. Method 2 (Forming the Composite Function).

16 **[Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials**

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $xe2$. Find the inverse function $f^{-1}(x)$.

Let $y = \frac{3x+1}{x-2} \Rightarrow y(x-2) = 3x + 1 \Rightarrow yx - 2y = 3x + 1 \Rightarrow x(y-3) = 2y + 1 \Rightarrow x = \frac{2y+1}{y-3}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-3}$. Final Answer: $f^{-1}(x) = \frac{2x+1}{x-3} \quad (xe3)$

17 **[Challenging] • Target: Function Mapping, Domain/Range and Inverses • Cross-Lesson: 4.01 Logarithmic and Exponential Functions**

A particle moves in a straight line with acceleration $a(t) = 6t - 12$ m/s^2. If it starts from rest at the origin (at $t = 0$), at what time $t > 0$ does the particle momentarily come to rest again?

Velocity $v(t) = \int (6t - 12)\, dt = 3t^2 - 12t + C$. Since $v(0) = 0 \Rightarrow C = 0$. Momentarily at rest when $v(t) = 0 \Rightarrow t(3t - 12) = 0 \Rightarrow t = \frac{12}{3}$ s. Final Answer: $t = \frac{12}{3}$ s

18 **[Challenging] • Target: Modulus Equations and Inequalities • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure**

In triangle ABC, the side lengths are $AB = (x + 2)$ cm, $BC = (3x - 1)$ cm, and $AC = 7$ cm. Given that $\angle ABC = 60^\circ$ and $x > 0$, find the exact area of triangle ABC.

Step 1: Use the cosine rule on triangle ABC to set up an equation for x:
$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(\angle ABC)$ Substitute the given values and $\cos(60^\circ) = \frac{1}{2}$:
$7^2 = (x+2)^2 + (3x-1)^2 - 2(x+2)(3x-1)\left(\frac{1}{2}\right)$ Step 2: Expand each term: $(x+2)^2 = x^2 + 4x + 4$
$(3x-1)^2 = 9x^2 - 6x + 1$ $(x+2)(3x-1) = 3x^2 + 5x - 2$ Substitute these back into the equation:
$49 = (x^2 + 4x + 4) + (9x^2 - 6x + 1) - (3x^2 + 5x - 2)$ $49 = 10x^2 - 2x + 5 - 3x^2 - 5x + 2$.

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19 **[Challenging]** • **Target: Quadratic Discriminant and Line-Parabola Intersections** • **Cross-Lesson: 6.01 Permutations and Combinations**

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2. During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm.

1. Express the surface area constraint: The total exterior surface area consists of the flat circular base, the curved cylindrical wall, and the hemispherical dome: $S = \pi r^2 + 2\pi rh + 2\pi r^2 = 3\pi r^2 + 2\pi rh$ Given $S = 180\pi$: $3\pi r^2 + 2\pi rh = 180\pi \Rightarrow 3r^2 + 2rh = 180$ 2. Find the value of r when $h = 16.5$ cm: $3r^2 + 2(16.5)r = 180 \Rightarrow 3r^2 + 33r - 180 = 0$ Dividing by 3: $r^2 + 11r - 60 = 0 \Rightarrow (r + 15)(r - 4) = 0$ Since $r > 0$, we have.

20 **[Challenging]** • **Target: Completing the Square and Applied Optimization** • **Cross-Lesson: 7.01 Differentiation and Rates of Change**

Two parallel lines, L_1 and L_2, have n and $n + 2$ distinct points marked on them respectively, where n is an integer with $n \geq 2$. Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear. Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is $6 : 5$, find the total number of triangles that can be formed.

1. Determine the number of triangles, T: Three points form a triangle if and only if they are not all on the same straight line. - Case 1: 1 point from L_1 and 2 points from L_2: $\binom{n}{1}\binom{n+2}{2} = n \cdot \frac{(n+2)(n+1)}{2}$ - Case 2: 2 points from L_1 and 1 point from L_2: $\binom{n}{2}\binom{n+2}{1} = \frac{n(n-1)}{2} \cdot (n + 2)$ Summing both cases: $T = \frac{n(n+2)}{2}[(n + 1) + (n - 1)] = \frac{n(n+2)}{2}(2n) = n^2(n + 2)$ 2. Determine the number of quadrilaterals, Q: To e.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|2x + 3| = 7$.

a. $x = 3$ and $x = -3$

b. $x = -2$ and $x = -\frac{10}{2}$

c. $x = 2$ only

d. **$x = 2$ and $x = -\frac{10}{2}$** ✓

Case 1: $2x + 3 = 7 \implies 2x = 4 \implies x = 2$. Case 2: $2x + 3 = -7 \implies 2x = -10 \implies x = -\frac{10}{2}$.

22 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|3x + 3| = 9$.

a. **$x = 2$ and $x = -\frac{12}{3}$** ✓

b. $x = 3$ and $x = -3$

c. $x = -2$ and $x = -\frac{12}{3}$

d. $x = 2$ only

Case 1: $3x + 3 = 9 \implies 3x = 6 \implies x = 2$. Case 2: $3x + 3 = -9 \implies 3x = -12 \implies x = -\frac{12}{3}$.

23 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|4x + 3| = 11$.

a. $x = 2$ only

b. $x = 3$ and $x = -3$

c. **$x = 2$ and $x = -\frac{14}{4}$** ✓

d. $x = -2$ and $x = -\frac{14}{4}$

Case 1: $4x + 3 = 11 \implies 4x = 8 \implies x = 2$. Case 2: $4x + 3 = -11 \implies 4x = -14 \implies x = -\frac{14}{4}$.

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24 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|5x + 3| = 13$.

a. $x = 2$ and $x = -\frac{16}{5}$ ✓

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{16}{5}$

d. $x = 3$ and $x = -3$

Case 1: $5x + 3 = 13 \implies 5x = 10 \implies x = 2$. Case 2: $5x + 3 = -13 \implies 5x = -16 \implies x = -\frac{16}{5}$.

25 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|6x + 3| = 15$.

a. $x = 3$ and $x = -3$

b. $x = 2$ and $x = -\frac{18}{6}$ ✓

c. $x = 2$ only

d. $x = -2$ and $x = -\frac{18}{6}$

Case 1: $6x + 3 = 15 \implies 6x = 12 \implies x = 2$. Case 2: $6x + 3 = -15 \implies 6x = -18 \implies x = -\frac{18}{6}$.

26 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|7x + 3| = 17$.

a. $x = 2$ and $x = -\frac{20}{7}$ ✓

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{20}{7}$

d. $x = 3$ and $x = -3$

Case 1: $7x + 3 = 17 \implies 7x = 14 \implies x = 2$. Case 2: $7x + 3 = -17 \implies 7x = -20 \implies x = -\frac{20}{7}$.

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27 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation \(|8x + 3| = 19 \).

a. \(x = 2 \) and \(x = -\frac{22}{8} \) ✓

b. \(x = 2 \) only

c. \(x = -2 \) and \(x = -\frac{22}{8} \)

d. \(x = 3 \) and \(x = -3 \)

Case 1: \(8x + 3 = 19 \implies 8x = 16 \implies x = 2 \). Case 2: \(8x + 3 = -19 \implies 8x = -22 \implies x = -\frac{22}{8} \).

28 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation \(|9x + 3| = 21 \).

a. \(x = 2 \) only

b. \(x = 2 \) and \(x = -\frac{24}{9} \) ✓

c. \(x = 3 \) and \(x = -3 \)

d. \(x = -2 \) and \(x = -\frac{24}{9} \)

Case 1: \(9x + 3 = 21 \implies 9x = 18 \implies x = 2 \). Case 2: \(9x + 3 = -21 \implies 9x = -24 \implies x = -\frac{24}{9} \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Function Mapping, Domain/Range and Inverses

The functions f and g are defined for $x \in \mathbf{R}$ by $f(x) = 2x - 5$ and $g(x) = x^2 + 3$. Find the composite value $(g \circ f)(4)$.

a. 12 ✓

b. 21

c. 14

d. 9

Evaluate $f(4) = 2(4) - 5 = 3$. Then $(g \circ f)(4) = g(3) = 3^2 + 3 = 12$.

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30 [Intermediate] • Target: Function Mapping, Domain/Range and Inverses

A function is defined by $f(x) = \frac{3x+1}{x-2}$ for $x \neq 2$. Find an expression for the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = \frac{2x+1}{x-3}, \quad x \neq 3$ ✓

b. $f^{-1}(x) = \frac{x-2}{3x+1}, \quad x \neq -\frac{1}{3}$

c. $f^{-1}(x) = \frac{2x-1}{x+3}, \quad x \neq -3$

d. $f^{-1}(x) = \frac{3x-2}{x-1}, \quad x \neq 1$

Let $y = \frac{3x+1}{x-2} \Rightarrow yx - 2y = 3x + 1 \Rightarrow x(y-3) = 2y + 1 \Rightarrow x = \frac{2y+1}{y-3}$. Thus $f^{-1}(x) = \frac{2x+1}{x-3}$ for $x \neq 3$.

31 [Intermediate] • Target: Function Mapping, Domain/Range and Inverses

The square root function $h(x) = 4 - \sqrt{2x - 6}$ is defined on its maximal real domain. Determine the domain and range of h.

a. **Domain:** $x \geq 3$**, Range:** $h(x) \leq 4$ ✓

b. Domain: $x \geq 3$, Range: $h(x) \geq 4$

c. Domain: $x > 3$, Range: $h(x) \in \mathbf{R}$

d. Domain: $x \leq 3$, Range: $h(x) \leq 0$

For the square root to exist: $2x - 6 \geq 0 \Rightarrow x \geq 3$. Since $\sqrt{2x-6} \geq 0$, $h(x) = 4 - \sqrt{2x-6} \leq 4$.

32 [Intermediate] • Target: Modulus Equations and Inequalities

Solve the linear modulus equation $|3x - 4| = 2x + 1$ for all valid real solutions.

a. $x = 5$ **and** $x = \frac{3}{5}$ ✓

b. $x = 5$ only

c. $x = -1$ and $x = 3$

d. No real solutions

Case 1: $3x - 4 = 2x + 1 \Rightarrow x = 5$ (valid). Case 2: $3x - 4 = -(2x + 1) \Rightarrow 5x = 3 \Rightarrow x = 3/5$ (valid).

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33 [Intermediate] • Target: Modulus Equations and Inequalities

Find the complete solution set for the absolute value inequality $|2x - 7| < 5$.

a. $1 < x < 6$ ✓

b. $x < 1$ or $x > 6$

c. $-6 < x < 1$

d. $2 < x < 5$

$-5 < 2x - 7 < 5 \Rightarrow 2 < 2x < 12 \Rightarrow 1 < x < 6$.

34 [Intermediate] • Target: Modulus Equations and Inequalities

Solve the two-modulus equation $|x + 2| = |2x - 5|$.

a. $x = 7$ **and** $x = 1$ ✓

b. $x = 7$ and $x = -1$

c. $x = 3$ and $x = -3$

d. $x = \frac{7}{3}$ only

$(x + 2)^2 = (2x - 5)^2 \Rightarrow x^2 + 4x + 4 = 4x^2 - 20x + 25 \Rightarrow 3x^2 - 24x + 21 = 0 \Rightarrow x = 7$ or $x = 1$.

35 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

Find the set of values of k for which the quadratic equation $x^2 + (k + 2)x + (2k + 1) = 0$ has two distinct real roots.

a. $k < 0$ **or** $k > 4$ ✓

b. $0 < k < 4$

c. $k \leq 0$ or $k \geq 4$

d. $-4 < k < 0$

$\Delta = (k + 2)^2 - 4(1)(2k + 1) = k^2 - 4k = k(k - 4) > 0 \Rightarrow k < 0$ or $k > 4$.

36 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

The line $y = 2x + c$ is tangent to the parabola $y = x^2 - 4x + 7$. Determine the exact value of the constant c.

a. $c = -2$ ✓

b. $c = 2$

c. $c = -7$

d. $c = 3$

$x^2 - 4x + 7 = 2x + c \Rightarrow x^2 - 6x + (7 - c) = 0$. Tangency requires $\Delta = 36 - 4(7 - c) = 8 + 4c = 0 \Rightarrow c = -2$.

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37 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

Find the range of values of m for which the line $y = mx - 1$ does not intersect the curve $y = 2x^2 + 3x + 1$.

a. $-1 < m < 7$ ✓

b. $m < -1$ or $m > 7$

c. $-7 < m < 1$

d. $m < 3$ or $m > 5$

$2x^2 + (3 - m)x + 2 = 0$. $\Delta = (3 - m)^2 - 16 = (m + 1)(m - 7) < 0 \Rightarrow -1 < m < 7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

By completing the square, express $f(x) = 2x^2 - 12x + 23$ in the form $a(x - h)^2 + k$, and state the minimum value of $f(x)$.

a. $2(x - 3)^2 + 5$**, minimum value** 5 ✓

b. $2(x - 3)^2 + 23$, minimum value 23

c. $2(x - 6)^2 - 13$, minimum value -13

d. $(2x - 6)^2 + 5$, minimum value 5

$2(x^2 - 6x) + 23 = 2[(x - 3)^2 - 9] + 23 = 2(x - 3)^2 + 5$. Minimum value is 5 at $x = 3$.

39 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 6.01 Permutations and Combinations

A rectangular field has three sides fenced with 80 m of fencing against a river. Find the maximum possible area.

a. $800\ \text{m}^2$ ✓

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

$A(x) = x(80 - 2x) = -2x^2 + 80x$. Vertex $x = 20$ m. Area $= 20(40) = 800\ \text{m}^2$.

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40 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find all real solutions to the biquadratic equation $x^4 - 10x^2 + 9 = 0$.

a. $x = \pm 1$ **and** $x = \pm 3$ ✓

b. $x = 1$ and $x = 9$

c. $x = \pm 1$ only

d. $x = \pm\sqrt{10}$

Let $u = x^2$: $(u - 1)(u - 9) = 0 \Rightarrow u = 1$ or $u = 9 \Rightarrow x = \pm 1, \pm 3$.

41 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The function $f(x) = (x + 11)^2 - 4$ has domain $x \geq -11$. State the range of f.

a. $f(x) \leq -4$

b. $f(x) \geq -4$ ✓

c. $f(x) \geq -11$

d. $x \geq -11$

Since $x \geq -11$, the minimum of f occurs at $x = -11$, giving $f(-11) = -4$, and f increases thereafter.

Range: $f(x) \geq -4$.

42 [Challenging] • Target: Function Mapping, Domain/Range and Inverses • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

The function $f(x) = (x + 8)^2 - 10$ has domain $x \geq -8$. State the range of f.

a. $f(x) \geq -8$

b. $f(x) \geq -10$ ✓

c. $f(x) \leq -10$

d. $x \geq -8$

Since $x \geq -8$, the minimum of f occurs at $x = -8$, giving $f(-8) = -10$, and f increases thereafter.

Range: $f(x) \geq -10$.

43 [Challenging] • Target: Modulus Equations and Inequalities • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The function $f(x) = (x + 7)^2 - 15$ has domain $x \geq -7$. State the range of f.

a. $f(x) \geq -7$

b. $x \geq -7$

c. $f(x) \leq -15$

d. $f(x) \geq -15$ ✓

Since $x \geq -7$, the minimum of f occurs at $x = -7$, giving $f(-7) = -15$, and f increases thereafter.

Range: $f(x) \geq -15$.

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44 [Challenging] • Target: Quadratic Discriminant and Line-Parabola Intersections • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

The function $f(x) = (x + 11)^2 + 5$ has domain $x \geq -11$. State the range of f.

a. $x \geq -11$

b. $f(x) \geq -11$

c. $f(x) \leq 5$

d. $f(x) \geq 5$ ✓

Since $x \geq -11$, the minimum of f occurs at $x = -11$, giving $f(-11) = 5$, and f increases thereafter. Range: $f(x) \geq 5$.

45 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The function $f(x) = (x + 6)^2 - 3$ has domain $x \geq -6$. State the range of f.

a. $f(x) \leq -3$

b. $f(x) \geq -3$ ✓

c. $f(x) \geq -6$

d. $x \geq -6$

Since $x \geq -6$, the minimum of f occurs at $x = -6$, giving $f(-6) = -3$, and f increases thereafter. Range: $f(x) \geq -3$.

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